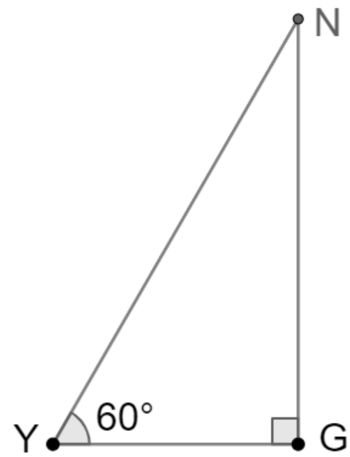
Geometry
Unit 5
Lesson 2 Practice

Name Answer Key________
Date___________________
Period__________________

Triangle NGY is shown where $m\angle Y = 60°$ and $m\angle G = 90°$.



Complete the steps in the construction of $\triangle HRT$ so that $\triangle HRT \cong \triangle NGY$ using two angles of $\triangle NGY$ and their included side.

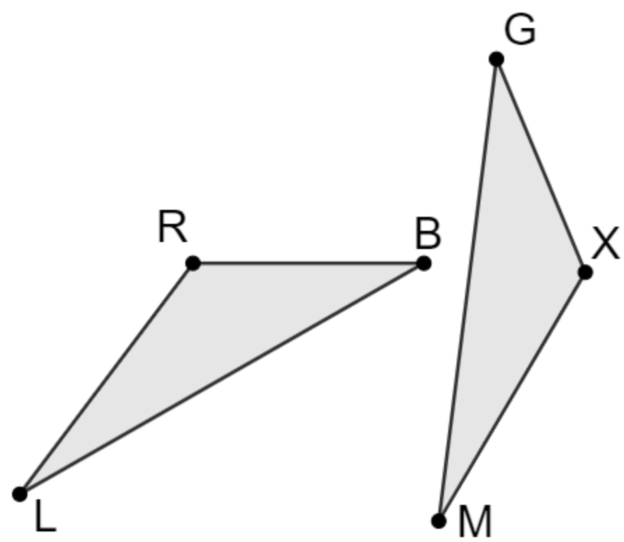
1.	Copy __$\overline{GY}$______ to construct $\overline{RT}$ so that ___$\overline{GY}$___ $\cong \overline{RT}$.
2.	Copy __$\angle Y$______ to construct $\angle T$ so that __$\angle Y$____ $\cong \angle T$.
3.	Copy __$\angle G$______ to construct $\angle R$ so that __$\angle G$____ $\cong \angle R$.
4.	Label the intersection of the sides of the two angles H.

5. Describe how to use geometric tools to determine if $\overline{YN} \cong \overline{TH}$, $\overline{GN} \cong \overline{RH}$, and $\angle N \cong \angle H$.
A straight edge, patty paper or ruler can be used to measure the lengths of each side to determine congruence. A protractor, compass or patty paper can be used to measure each angle and determine congruence.

6. What postulate or theorem does the construction illustrate?

Angle-Side-Angle (ASA) Congruence

Triangles RBL and GXM are shown.



Identify if the given information is sufficient to determine whether the triangles are congruent. If so, state the congruence theorem or postulate.

	Given Information	Are the Triangles Congruent?	Theorem or Postulate Used
7.	$\overline{XG} \cong \overline{RB}, \angle RBL \cong \angle XGM$, $\overline{BL} \cong \overline{GM}$	Yes	SAS
8.	$\angle BLR \cong \angle XMG$, $\overline{RL} \cong \overline{XM}$, $\angle BRL \cong \angle GXM$	Yes	ASA
9.	$\overline{BL} \cong \overline{GM}$, $\angle BLR \cong \angle XMG$, $\overline{XG} \cong \overline{RB}$	No	
10.	$\overline{XG} \cong \overline{RB}$, $\overline{BL} \cong \overline{GM}$, $\overline{RL} \cong \overline{XM}$	Yes	SSS